

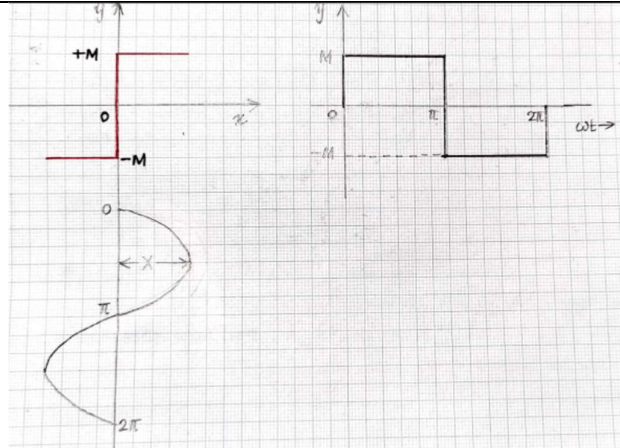
Scheme of Valuation/Answer Key			
(Scheme of evaluation (marks in brackets) and answers of problems/key)			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
SEVENTH SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2022			
Course Code: EET401			
Course Name: ADVANCED CONTROL SYSTEMS			
Max. Marks: 100			Duration: 3 Hours
PART A			
		Answer all questions, each carries 3 marks.	Marks
1		(i) state variables (1mark) (ii) phase variables (1mark) and (iii) state space (1mark)	(3)
2		Select state variables and output variable, substitute them in the basic Kirchoff's equations and rearrange the terms. (1 mark) $v(t) = Ri(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int i(t)dt = Ri(t) + L \frac{di(t)}{dt} + v_c(t)$ Select voltage across capacitor as one state variable and inductor current (current in the series circuit) as the other state variable. Rearrange the terms. Final state equation (1 mark) $\dot{v}_o(t) = Ri(t)$ Substitute for $i(t)$ in terms of state variable. Final output equation (1 mark)	(3)
3		Statement of property and proof 1 mark for each property	(3)
4		$m = 5$ Method with steps 2 marks and 1 mark for final answer	(3)
5		Duality principle- Explanation with necessary equations (3 marks)	(3)
6		Explanation with necessary equations: New state model (2 marks) and condition (1mark) $\dot{Z} = \Lambda Z + \tilde{B}U$ where $\Lambda = M^{-1}AM$ & $\tilde{B} = M^{-1}B$ (2 marks) Condition: $\tilde{B}$ must have no rows with all zeros (1mark)	(3)
7		Any three characteristics (without explanation) (1 mark each) Jump resonance, Limit cycle, Frequency entrainment, Sub-harmonic oscillations, etc.	(3)
8		(i) jump resonance (1mark) (ii) stable limit cycle (1mark) and (iii) unstable limit cycle (1mark)	(3)
9		(i) singular point (1mark) (ii) vortex point (1mark) and (iii) saddle point (1mark)	(3)

10		(i) positive definite & example (1 mark) (ii) negative definite & example (1 mark) and (iii) indefinite & example (1 mark)	(3)
PART B			
<i>Answer one full question from each module, each carries 14 marks.</i>			
		Module I	
11	a)	State model of an armature controlled dc motor: KVL equation for armature circuit: (1 mark) $V_a = I_a R_a + L_a \frac{di_a}{dt} + E_b \quad \left(E_b \propto \frac{d\theta}{dt} \right)$ Torque equation: $T \propto I_a$ ($T = K_T I_a$) (1 mark) Eqn. for electro mechanical system: $J \frac{d^2\theta}{dt^2} + B \frac{d\theta}{dt} = T$ (1 mark) Select suitable state variables and output variables, substitute them in the above three equations, rearrange the terms. (2 marks) Final state equation (1 mark) Final output equation (1 mark)	(7)
	b)	State model using phase variables $x_1 = y$ (1 mark) $x_2 = \frac{dy}{dt} = \dot{x}_1$ (1 mark) $x_3 = \frac{d^2y}{dt^2} = \dot{x}_2$ (1 mark) $\therefore \frac{d^3y}{dt^3} + 6 \frac{d^2y}{dt^2} + 11 \frac{dy}{dt} + 6y + u = 0 \Rightarrow \dot{x}_3 = -6x_3 - 11x_2 - 6x_1 + u$ (2 marks) $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} u \quad [y] = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ (2 marks)	(7)
		OR	
12	a)	Similarity transformation: (3 marks) Process of transforming a square matrix to another similar matrix by a transformation. Similarity transformation in state space analysis: (2 marks) The helpfulness of similarity transformation are, a. Diagonalization of a square matrix - Diagonal Canonical Form	(5)

		b. If we transform the system to controllable canonical form, state feedback controller design is easier c. If we transform the system to observable canonical form, observer design is easier d. If we transform the system to Diagonal/Jordan canonical form controllability, observability and stability can be easily identified, solution (STM) can be computed easily Any one point from the above is sufficient.	
	b)	Canonical form $[\lambda I - A] = \begin{bmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 6 & 11 & \lambda + 6 \end{bmatrix} \quad \lambda_1 = -1 \quad \lambda_2 = -2 \quad \lambda_3 = -3$ <i>(3 marks)</i> $M = [m_1 \quad m_2 \quad m_3] = \begin{bmatrix} 6 & 3 & 2 \\ -6 & -6 & -6 \\ 6 & 12 & 18 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix} (3 marks) A_{new} = M^{-1} . A . M = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \quad B_{new} = M^{-1} . B = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} (3 marks) C_{new} = C . M = [1 \quad 1 \quad 1] $	(9)
		Module II	
13		$z^2 Y(z) + 5zY(z) + 6Y(z) = U(z) \quad \therefore \frac{Y(z)}{U(z)} = \frac{1}{(z+3)(z+2)}$ <i>(2 marks)</i> On solving, $\frac{Y(z)}{U(z)} = \frac{-1}{(z+3)} + \frac{1}{(z+2)}$ <i>(4 marks)</i> $x_1(k+1) = -3x_1(k) + u(k)$ $x_2(k+1) = -2x_2(k) + u(k)$ $y(k) = -x_1(k) + x_2(k)$ $\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \end{bmatrix} = \begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} u(k) \\ u(k) \end{bmatrix} \quad \text{50\% credit for steps}$ $y(k) = \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix}$ <i>(6 marks)</i> Block diagram <i>(1 mark)</i>	(14)

		STM: $A^k = Z^{-1} \left\{ [zI - A]^{-1} . z \right\}$ $= Z^{-1} \left\{ \begin{bmatrix} \frac{1}{(z+3)} & 0 \\ 0 & \frac{1}{(z+2)} \end{bmatrix} z \right\}$ $= \begin{bmatrix} -3^k & 0 \\ 0 & -2^k \end{bmatrix}$ (1 mark) 50% credit for steps The STM can be directly written without above computation also.	
		OR	
14	a)	STM using Laplace transform method $[sI - A] = \begin{bmatrix} (s-1) & 0 \\ -1 & (s-1) \end{bmatrix}$ $STM = e^{At} = L^{-1} \left\{ [sI - A]^{-1} \right\}$ $= \begin{bmatrix} e^t & 0 \\ te^t & e^t \end{bmatrix}$ $X(t) = e^{At} . X(0)$ $= \begin{bmatrix} e^t \\ te^t \end{bmatrix}$ (1 mark) (1 mark) (3 marks) 50% credit for steps (1 mark) (1 mark)	(7)
	b)	STM using Cayley-Hamilton Theorem $ \lambda I - A = \begin{vmatrix} \lambda & -1 \\ 2 & \lambda + 3 \end{vmatrix} \quad \lambda_1 = -1 \text{ \& } \lambda_2 = -2$ $f(\lambda_1) = \alpha_0 + \alpha_1 . \lambda_1 = e^{\lambda_1 t} = e^{-t}$ $f(\lambda_2) = \alpha_0 + \alpha_1 . \lambda_2 = e^{\lambda_2 t} = e^{-2t}$ $\therefore \alpha_0 = 2e^{-t} - e^{-2t} \quad \& \quad \alpha_1 = e^{-t} - e^{-2t}$ $f(A) = e^{At} = \alpha_0 . I + \alpha_1 . A = \begin{bmatrix} (2e^{-t} - e^{-2t}) & (e^{-t} - e^{-2t}) \\ (-2e^{-t} + 2e^{-2t}) & (-e^{-t} + 2e^{-2t}) \end{bmatrix}$ (1 mark) (1 mark) (1 mark) (2 marks) (2 marks)	(7)
		Module III	
15	a)	Kalman's method for observability Composite matrix : $Q_0 = \begin{bmatrix} C^T & A^T . C^T & (A^T)^2 . C^T \end{bmatrix}$ (2 marks)	(6)

		$Q_0 = \begin{bmatrix} 40 & 0 & 0 \\ 10 & 40 & -30 \\ 0 & 10 & 0 \end{bmatrix}$ <i>50% credit for steps</i> (3 marks) $Q_0 \neq 0$ System is completely observable (1 mark)	
	b)	<i>There is no method specified for design in the question. Examiners are requested to verify the method applied and award the marks accordingly. 7 marks for steps.</i> Observer gain matrix $G = \begin{bmatrix} 25.2 \\ 21.5 \\ 10 \end{bmatrix}$ (1 mark)	(8)
		OR	
16	a)	PBH method for controllability : condition $\text{rank} \left\{ \begin{bmatrix} (\lambda I - A) & B \end{bmatrix} \right\} = n$ (1 mark) $n = 2 \quad \lambda_1 = -1 \quad \lambda_2 = -4$ (2 marks) For λ_1 : $\text{rank} \left\{ \begin{bmatrix} -1 & -1 & 0 \\ 4 & 4 & 1 \end{bmatrix} \right\} = 2$ (1 mark) For λ_2 : $\text{rank} \left\{ \begin{bmatrix} -4 & -1 & 0 \\ 4 & 1 & 1 \end{bmatrix} \right\} = 2$ (1 mark) System is completely controllable. (1 mark)	(6)
	b)	Feed back controller with a state feed back: <i>There is no method specified for design in the question. Examiners are requested to verify the method applied and award the marks accordingly. 7 marks for steps.</i> State feedback gain matrix: $K = [0.4 \quad 0.4 \quad 0.1]$ (1 mark)	(8)
		Module IV	
17	a)	Describing function of ideal relay non-linearity: Waveforms :	(6)



(1 mark)

Input signal: $x = X \sin \omega t$

Output signal:

$$y(\omega t) = \begin{cases} +M & ; 0 < \omega t < \pi \\ -M & ; \pi < \omega t < 2\pi \end{cases}$$

(1 mark)

Describing function :

$$A_1 = \frac{4M}{\pi} \quad B_1 = 0 \quad \phi_1 = 0^\circ$$

(2 marks)

Derivation of constants with necessary steps

$$K_N(X, \omega) = \frac{Y_1}{X} \angle \phi_1 \quad Y_1 = \sqrt{A_1^2 + B_1^2} \quad \phi_1 = \tan^{-1} \left(\frac{B_1}{A_1} \right)$$

(2 marks)

$$K_N(X, \omega) = \frac{4M}{\pi X} \angle 0^\circ$$

b) Frequency and nature of the limit cycle:

(8)

Plot of $G(j\omega)$

(2 marks)

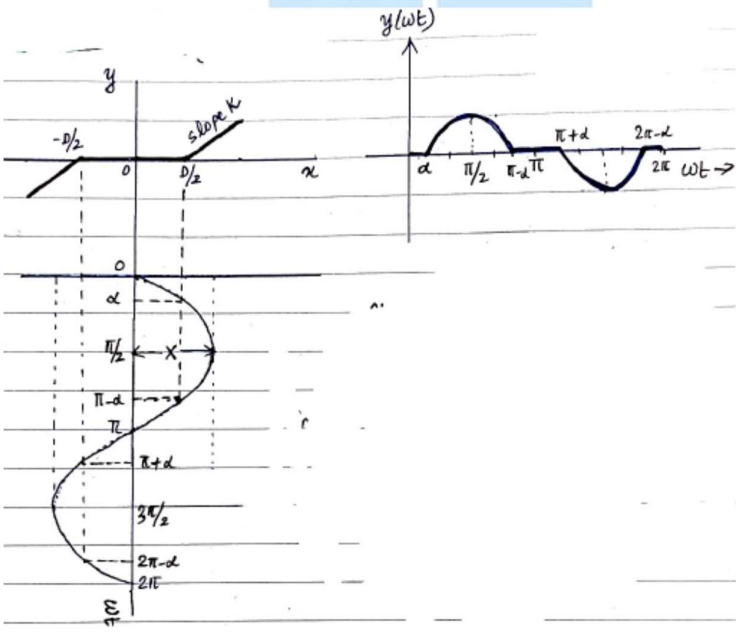
$$\frac{-1}{K_N(X, \omega)} \text{ locus}$$

(2 marks)

$$G(j\omega) = \frac{10}{s(s+1)(s+2)} = \frac{5}{j\omega(1+j\omega)(1+j0.5\omega)}$$

$$|G(j\omega)| = \frac{5}{\omega \sqrt{(1+\omega^2)} \sqrt{(1+0.25\omega^2)}} \quad \angle G(j\omega) = -90^\circ - \tan^{-1}(\omega) - \tan^{-1}(0.5\omega)$$

$$\frac{-1}{K_N(X, \omega)} = \frac{\pi X}{4} \angle -180^\circ$$

		<table border="1"> <tr> <td>ω rad/s</td><td>.....</td><td>.....</td><td>.....</td></tr> <tr> <td>$G(j\omega)$</td><td>.....</td><td>.....</td><td>.....</td></tr> <tr> <td>$\angle G(j\omega)$</td><td>.....</td><td>-180^0</td><td>.....</td></tr> </table>	ω rad/s				$ G(j\omega) $				$\angle G(j\omega)$		-180^0		
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		$G(j\omega)$ locus and $\frac{-1}{K_N(X, \omega)}$ locus intersect at -180^0 axis (1 mark) $-90 - \tan^{-1}(\omega) - \tan^{-1}(0.5\omega) = -180^0$ (2 marks) $\therefore \omega = \sqrt{2} \text{ rad / s}$ Nature of the limit cycle: stable (1 mark) <i>Student may assume the shape of $G(j\omega)$ as it crosses the other locus at -180^0 axis. In that case, they may only need to calculate the point of intersection.</i>													
		OR													
18	a)	Describing function of dead-zone non-linearity: Waveforms :  Input signal: $x = X \cdot \sin \omega t$ Output signal: $y(\omega t) = \begin{cases} 0 & ; 0 < \omega t < \alpha \\ K \left(x - \frac{D}{2} \right) & ; \alpha < \omega t < (\pi - \alpha) \\ 0 & ; (\pi - \alpha) < \omega t < \pi \end{cases}$	(6)												

	Describing function : $K_N(X, \omega) = \frac{Y_1}{X} \angle \phi_1 \quad Y_1 = \sqrt{A_1^2 + B_1^2} \quad \phi_1 = \tan^{-1} \left(\frac{B_1}{A_1} \right) \quad (2 \text{ marks})$ $A_1 = \frac{4K}{\pi} \left[\frac{X}{2} \left(\frac{\pi}{2} - \alpha + \frac{\sin 2\alpha}{2} \right) - \frac{D}{2} \cos \alpha \right] = KX \left[1 - \frac{2}{\pi} (\alpha + \sin \alpha \cdot \cos \alpha) \right]$ $B_1 = 0 \quad \phi_1 = 0^\circ$ Derivation of constants with necessary steps (final answer either in terms of D or without D) (2 marks) <i>Alternate expression for A_1: $A_1 = \frac{KX}{\pi} \{\pi - 2\alpha - \sin 2\alpha\}$</i>													
b)	Frequency and nature of the limit cycle: Plot of $G(j\omega)$ (2 marks) $\frac{-1}{K_N(X, \omega)}$ locus (2 marks) $G(j\omega) = \frac{1}{j\omega(1+j2\omega)(1+j\omega)}$ $ G(j\omega) = \frac{1}{\omega \sqrt{(1+4\omega^2)} \sqrt{(1+\omega^2)}} \quad \angle G(j\omega) = -90^\circ - \tan^{-1}(2\omega) - \tan^{-1}(\omega)$ $\frac{-1}{K_N(X, \omega)} = \begin{cases} \frac{\pi}{2K\pi[\beta + \sin \beta \cdot \cos \beta]} \angle -180^\circ & ; X > 1 \\ \frac{1}{K} \angle -180^\circ & ; X < 1 \end{cases}$ <table border="1" style="width: 100%; text-align: center; border-collapse: collapse;"> <tr> <td>ω rad/s</td> <td>.....</td> <td>0.707</td> <td>.....</td> </tr> <tr> <td>$G(j\omega)$</td> <td>.....</td> <td>.....</td> <td>.....</td> </tr> <tr> <td>$\angle G(j\omega)$</td> <td>.....</td> <td>-180⁰</td> <td>.....</td> </tr> </table> $G(j\omega)$ locus and $\frac{-1}{K_N(X, \omega)}$ locus intersect at -180⁰ axis (1 mark) $-90 - \tan^{-1}(2\omega) - \tan^{-1}(\omega) = -180^\circ$ $\therefore \omega = 0.707 \text{ rad/s}$ (2 marks) Nature of the limit cycle: stable (1 mark) <i>Student may assume the shape of $G(j\omega)$ as it crosses the other locus at -180⁰ axis. In that case, they may only need to calculate the point of intersection.</i>	ω rad/s		0.707		$ G(j\omega) $				$\angle G(j\omega)$		-180 ⁰		(8)
ω rad/s		0.707												
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$\angle G(j\omega)$		-180 ⁰												

		Module V	
19	a)	Phase trajectory (1 mark) Procedure to draw the phase trajectory with necessary sketches (6 marks)	(7)
	b)	Lyapunov function: $-Q = A^T \cdot P + P \cdot A$ $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix}^T \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} + \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix} \quad (p_{12} = p_{21})$ (1 mark) On solving, $P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} = \begin{bmatrix} 23/60 & -7/60 \\ -7/60 & 11/60 \end{bmatrix}$ (3 marks) Positive definiteness of P for stability. $ p_{11} > 0$ $\begin{vmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{vmatrix} = \frac{17}{300} > 0$ (1 mark) Lyapunov function: $V(X) = X^T \cdot P \cdot X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}^T \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{23}{60} x_1^2 - \frac{14}{60} x_1 \cdot x_2 + \frac{11}{60} x_2^2$ (2 marks)	(7)
		OR	
20	a)	Plot of phase trajectory (with necessary steps) (7 marks) full marks may be awarded if the shape of trajectory is identified with atleast 3 points marked and joined by a smooth curve (including the initial point).	(7)
	b)	Lyapunov function (1 mark) Condition/s for the system to be stable (1 mark) Condition/s for the system to be asymptotically stable in large (1 mark) Condition/s for the system to be unstable (1 mark) $\frac{dV(X)}{dt} = \left(\frac{\partial V}{\partial x_1} \right) \dot{x}_1 + \left(\frac{\partial V}{\partial x_2} \right) \dot{x}_2 = -2x_2^4$ (2 marks) System is stable (1 mark)	(7)
